

Differentiation · Worksheet 1

Chapter 7: Differentiation. Answer all questions. Show your working clearly. Total: 273 marks. Time: 5 hours 40 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are thirteen sections, one for each topic in Chapter 7. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- All angles are in **radians** unless stated otherwise.
- Give non-exact answers to 3 significant figures unless stated otherwise.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Differentiation from first principles	19	
B Differentiating polynomials; tangents and normals	22	
C Stationary points; increasing and decreasing functions	22	
D Points of inflection, convexity and concavity	21	
E Differentiating exponentials and logarithms	19	
F Differentiating trigonometric functions	20	
G The chain rule	19	
H The product rule	19	
I The quotient rule	21	
J Implicit differentiation	20	
K Parametric differentiation	21	
L Connected rates of change	20	
M Optimisation problems	26	
Roar-level challenge	4	
Total	273	



Section A Differentiation from first principles

19 marks

1. The points $P(3, 9)$ and $Q(3 + h, (3 + h)^2)$ lie on the curve $y = x^2$. [3]

(a) Show that the gradient of the chord PQ is $6 + h$.

(b) Hence state the gradient of the curve at P , explaining your reasoning.

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2. Prove, from first principles, that the derivative of $4x^2$ is $8x$. [3]

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3. Prove, from first principles, that if $f(x) = x^2 - 5x + 2$, then $f'(x) = 2x - 5$. [4]

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4. Prove, from first principles, that the derivative of x^3 is $3x^2$.

[4]

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5. Given that

[5]

$$\lim_{h \rightarrow 0} \frac{\sin h}{h} = 1 \quad \text{and} \quad \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} = 0,$$

prove, from first principles, that the derivative of $\sin x$ is $\cos x$.

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Section B Differentiating polynomials; tangents and normals

22 marks

6. Find $\frac{dy}{dx}$ when $y = 4x^3 - 5x^2 + 7x - 2$. [2]

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7. Find $\frac{dy}{dx}$ when $y = 6\sqrt{x} + \frac{8}{x^2} - \frac{1}{3x}$. [3]

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8. Find the gradient of the curve $y = \frac{x^2 + 3x}{\sqrt{x}}$ at the point where $x = 4$. [3]

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9. Find an equation of the normal to the curve $y = x^3 - 2x + 1$ at the point where $x = 2$. [4]
Give your answer in the form $ax + by + c = 0$, where a , b and c are integers.

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10. The curve C has equation $y = x^3 - 6x^2 + 9x + 1$. The point P on C has x -coordinate 4.

(a) Find $\frac{dy}{dx}$. [2]

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(b) Find an equation of the tangent to C at P . [3]

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(c) The tangent to C at the point Q is parallel to the tangent at P . [3]
Find the coordinates of Q .

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(d) Find an equation of the normal to C at Q . [2]

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(Total for Question 10 is 10 marks)



Section C Stationary points; increasing and decreasing functions

22 marks

11. Find the coordinates of the stationary point on the curve $y = x^2 - 6x + 4$, and state whether it is a maximum or a minimum. [3]

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12. Find the coordinates of the stationary points on the curve $y = x^3 - 3x^2 - 9x + 5$, and use the second derivative to determine their nature. [5]

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13. Find the set of values of x for which $f(x) = x^3 - 12x$ is an increasing function. [3]

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14. Prove that $f(x) = x^3 + 3x^2 + 5x - 1$ is an increasing function for all real values of x . [3]

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15. The curve C has equation

$$y = 2x + \frac{8}{x^2}, \quad x > 0.$$

- (a) Find the coordinates of the stationary point on C . [4]

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- (b) Use $\frac{d^2y}{dx^2}$ to determine the nature of this stationary point. [2]

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- (c) Hence state the set of values of k for which the line $y = k$ meets C at two distinct points. [2]

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(Total for Question 15 is 8 marks)



Section D Points of inflection, convexity and concavity

21 marks

16. Given $f(x) = x^4 - 2x^3 + 5$, find $f'''(x)$. [2]

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17. Find the coordinates of the point of inflection on the curve $y = x^3 - 6x^2 + 5$, showing that it is a genuine point of inflection. [3]

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18. The curve C has equation $y = x^3 - 3x^2 + 4$. [3]
Find the set of values of x for which C is concave, and the set of values for which C is convex.

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19. The curve C has equation $y = x^4 - 4x^3$.

[5]

Find the coordinates of the two points of inflection on C , and determine which one is a stationary point of inflection.

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20. $f(x) = 3x^5 - 5x^4$

- (a) Find $f''(x)$ and show that $f''(x) = 0$ when $x = 0$ and when $x = 1$. [3]

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- (b) Show that the curve $y = f(x)$ has a point of inflection at $x = 1$, and find its coordinates. [3]

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- (c) Explain why the curve does **not** have a point of inflection at $x = 0$, even though $f''(0) = 0$. [2]

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(Total for Question 20 is 8 marks)



Section E Differentiating exponentials and logarithms

19 marks

21. Find $\frac{dy}{dx}$ when $y = 3e^x + 4\ln x$. [2]

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22. Find $\frac{dy}{dx}$ when $y = e^{5x} - \ln(2x)$. [2]

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23. Given $y = 2^x$, find $\frac{dy}{dx}$, and hence find the exact gradient of the curve $y = 2^x$ at the point where $x = 3$. [3]

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24. Find the exact coordinates of the stationary point on the curve $y = e^{2x} - 6x$, and determine its nature. [4]

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25. The curve C has equation $y = 2 \ln x - x + 5$, $x > 0$.

(a) Find the exact coordinates of the stationary point on C . [3]

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(b) Determine the nature of this stationary point. [2]

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(c) Find an equation of the tangent to C at the point where $x = 1$. [3]

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(Total for Question 25 is 8 marks)



Section F Differentiating trigonometric functions

20 marks

26. Find $\frac{dy}{dx}$ when $y = 3 \sin x - 2 \cos x$. [2]

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27. Find $\frac{dy}{dx}$ when $y = \sin 4x + \cos\left(\frac{x}{2}\right)$. [2]

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28. Find the gradient of the curve $y = 5 \tan x - 2x$ at the point where $x = \frac{\pi}{4}$. [3]

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29. Find the exact coordinates of the stationary points on the curve $y = x + 2 \cos x$, $0 \leq x \leq 2\pi$, and determine their nature. [5]

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30. The curve C has equation $y = 2 \sin x + \sin 2x$, $0 \leq x \leq 2\pi$.

(a) Find $\frac{dy}{dx}$. [2]

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(b) Hence find the x -coordinates of the stationary points on C . [4]

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(c) Find an equation of the tangent to C at the origin. [2]

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(Total for Question 30 is 8 marks)



Section G The chain rule

19 marks

31. Find $\frac{dy}{dx}$ when $y = (3x + 1)^5$. [2]

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32. Given $y = \sqrt{x^2 + 5}$, find $\frac{dy}{dx}$, and find the gradient of the curve at the point where $x = 2$. [3]

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33. Differentiate with respect to x : (a) e^{x^2} , (b) $\ln(x^2 + 1)$. [3]

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34. Differentiate with respect to x : (a) $\sin^3 x$, (b) $\cos(x^3)$. [3]

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35. The curve C has equation $y = \sqrt{4x+1}$.

(a) Find $\frac{dy}{dx}$. [2]

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(b) Find an equation of the normal to C at the point where $x = 2$, in the form $ax + by + c = 0$. [3]

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(c) The normal meets the x -axis at A and the y -axis at B . Find the area of triangle OAB , where O is the origin. [3]

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(Total for Question 35 is 8 marks)



Section H The product rule

19 marks

36. Find $\frac{dy}{dx}$ when $y = x^2 e^x$. [2]

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37. Find $\frac{dy}{dx}$ when $y = x^3 \sin x$. [2]

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38. Find the gradient of the curve $y = (x + 1) \ln x$ at the point where $x = 1$. [3]

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39. Given $y = x(2x - 1)^4$, show that $\frac{dy}{dx} = (2x - 1)^3(ax + b)$, where a and b are integers to be found. [4]

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40. $f(x) = x^2e^{-x}$

(a) Show that $f'(x) = e^{-x}(2x - x^2)$.

[3]

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(b) Find the exact coordinates of the stationary points of $y = f(x)$.

[3]

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(c) Given that $f''(x) = e^{-x}(x^2 - 4x + 2)$, determine the nature of each stationary point.

[2]

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(Total for Question 40 is 8 marks)



Section I The quotient rule

21 marks

41. Find $\frac{dy}{dx}$ when $y = \frac{x}{x+1}$, simplifying your answer. [2]

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42. Find $\frac{dy}{dx}$ when $y = \frac{e^{2x}}{x}$, giving your answer as a single fraction. [3]

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43. By writing $\tan x = \frac{\sin x}{\cos x}$, use the quotient rule to prove that $\frac{d}{dx}(\tan x) = \sec^2 x$. [3]

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44. Find the coordinates of the stationary points on the curve

[5]

$$y = \frac{x^2 + 3}{x - 1}, \quad x \neq 1.$$

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45. $f(x) = \frac{\ln x}{x}$, $x > 0$.

(a) Show that $f'(x) = \frac{1 - \ln x}{x^2}$. [3]

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(b) Hence find the exact coordinates of the stationary point of $y = f(x)$. [2]

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(c) Show that $f''(x) = \frac{2 \ln x - 3}{x^3}$, and hence determine the nature of the stationary point. [3]

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(Total for Question 45 is 8 marks)



Section J Implicit differentiation

20 marks

46. Find $\frac{dy}{dx}$ in terms of x and y for the curve $x^2 + y^2 = 25$. [2]

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47. Find $\frac{dy}{dx}$ in terms of x and y for the curve $x^2 + 3xy + y^2 = 11$. [3]

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48. Find the gradient of the curve $x^3 + y^3 = 9$ at the point $(2, 1)$. [3]

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49. The curve C has equation $x^2 - xy + y^2 = 7$. [4]
Show that the point $(3, 2)$ lies on C , and find an equation of the tangent to C at this point.

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50. The curve C has equation

$$x^2 + 2xy - 3y^2 + 16 = 0.$$

(a) Find $\frac{dy}{dx}$ in terms of x and y .

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(b) Find the coordinates of the points on C where $\frac{dy}{dx} = 0$.

[4]

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(Total for Question 50 is 8 marks)



Section K Parametric differentiation

21 marks

51. A curve has parametric equations $x = t^2$, $y = 4t$. Find $\frac{dy}{dx}$ in terms of t . [2]

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52. A curve has parametric equations $x = 3 \cos \theta$, $y = 2 \sin \theta$. [3]
Find $\frac{dy}{dx}$ in terms of θ , and its value when $\theta = \frac{\pi}{4}$.

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53. A curve has parametric equations $x = t + \frac{1}{t}$, $y = t - \frac{1}{t}$, $t > 0$. [3]
Find the gradient of the curve at the point where $t = 2$.

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54. A curve has parametric equations $x = 2t^2$, $y = 4t$. [4]
Find an equation of the tangent to the curve at the point where $t = 3$, in the form $ax + by + c = 0$.

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55. The curve C has parametric equations

$$x = t^2 - 1, \quad y = t^3 - 3t.$$

- (a) Find $\frac{dy}{dx}$ in terms of t . [3]

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- (b) Find the coordinates of the stationary points on C . [3]

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- (c) Find an equation of the normal to C at the point where $t = 2$, in the form $ax + by + c = 0$. [3]

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(Total for Question 55 is 9 marks)



Section L Connected rates of change

20 marks

56. Given $y = x^3$ and $\frac{dx}{dt} = 2$, find $\frac{dy}{dt}$ when $x = 3$. [2]

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57. The area of a circular ink blot is increasing at a rate of $10 \text{ cm}^2 \text{ s}^{-1}$. [3]
Find the rate at which the radius is increasing when the radius is 5 cm.

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58. The edges of a cube are increasing at a rate of 0.5 cm s^{-1} . [3]
Find the rate at which the volume is increasing when each edge is 4 cm long.

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59. A spherical balloon is inflated so that its volume increases at a constant rate of $50 \text{ cm}^3 \text{ s}^{-1}$. [4]
Find the rate at which the radius is increasing when the radius is 5 cm.
[Volume of a sphere = $\frac{4}{3}\pi r^3$]

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60. Water is poured into an inverted right circular cone at a constant rate of $8 \text{ cm}^3 \text{ s}^{-1}$.
When the depth of the water is h cm, the radius of the water surface is $\frac{1}{2}h$ cm.

(a) Show that the volume of water in the cone is $V = \frac{\pi h^3}{12}$. [Volume of a cone = $\frac{1}{3}\pi r^2 h$] [2]

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(b) Find the rate at which the depth of water is increasing when $h = 4$. [4]

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(c) Find the rate at which the radius of the water surface is increasing at this instant. [2]

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(Total for Question 60 is 8 marks)



Section M Optimisation problems

26 marks

61. Two positive numbers have a sum of 20. Use calculus to find the maximum possible value of their product. [3]

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62. A square sheet of card has sides of length 12 cm. Squares of side x cm are cut from each corner, and the sides are folded up to make an open box. [5]

(a) Show that the volume of the box is $V = x(12 - 2x)^2$.

(b) Find the value of x that maximises the volume, and the maximum volume.

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63. A closed cylindrical can has radius r cm and height h cm, and a volume of 250π cm³. [5]

(a) Show that the total surface area is $S = 2\pi r^2 + \frac{500\pi}{r}$.

(b) Find the minimum surface area, giving your answer as an exact multiple of π .

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64. A farmer has 60 m of fencing to make a rectangular enclosure against a long straight wall, so only three sides need fencing. [4]

Find the maximum area that can be enclosed.

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65. A cuboid has a rectangular base of width x cm and length $2x$ cm, and height h cm.
The total surface area of the cuboid is 600 cm^2 .

(a) Show that the volume, $V \text{ cm}^3$, of the cuboid is given by [3]

$$V = 200x - \frac{4}{3}x^3.$$

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(b) Use calculus to find the exact value of x for which V is stationary, and find the maximum volume, [4]
to 3 significant figures.

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(c) Justify that this value of V is a maximum. [2]

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(Total for Question 65 is 9 marks)



ROAR-LEVEL CHALLENGE

4 marks

66. By taking logarithms, find the exact coordinates of the stationary point on the curve $y = x^x$, $x > 0$. [4]

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END OF WORKSHEET

Total: 273 marks